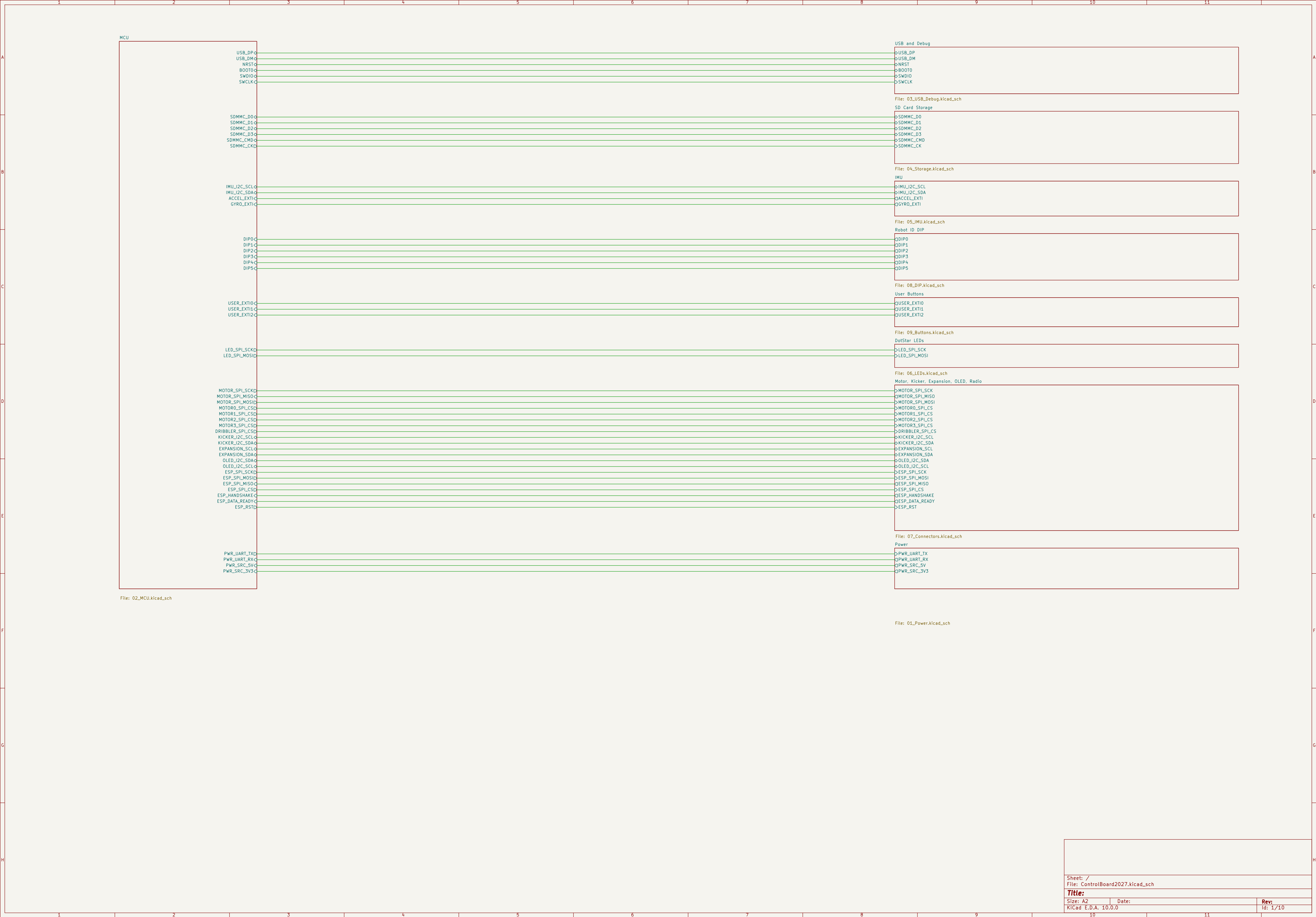
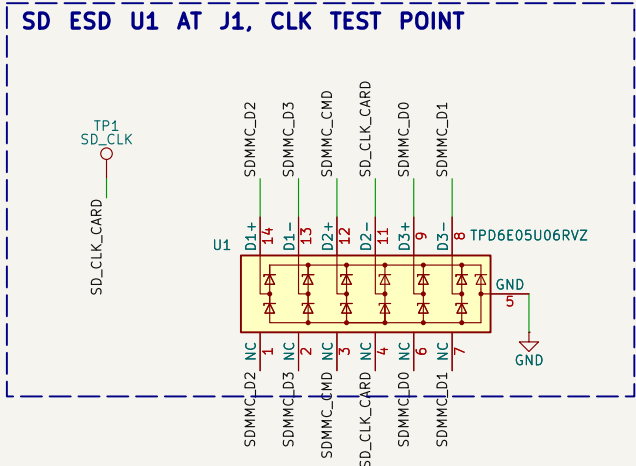
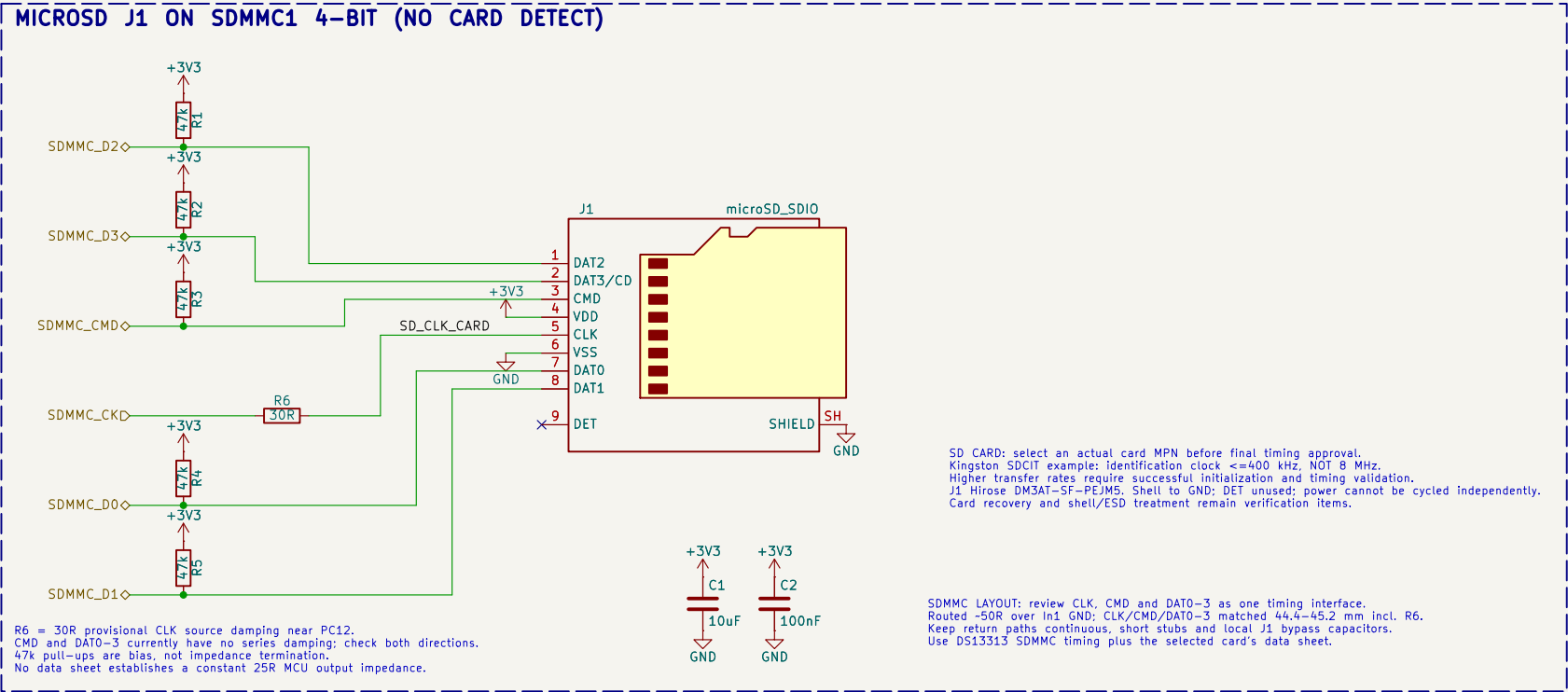
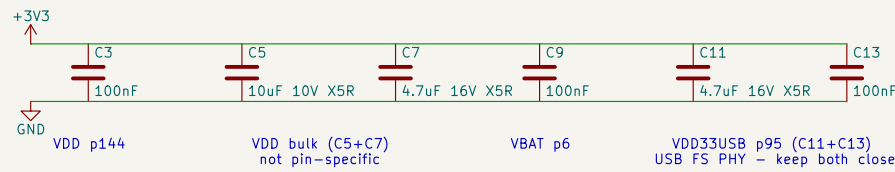
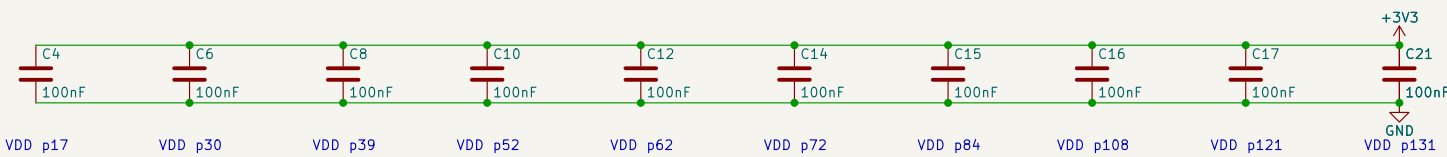






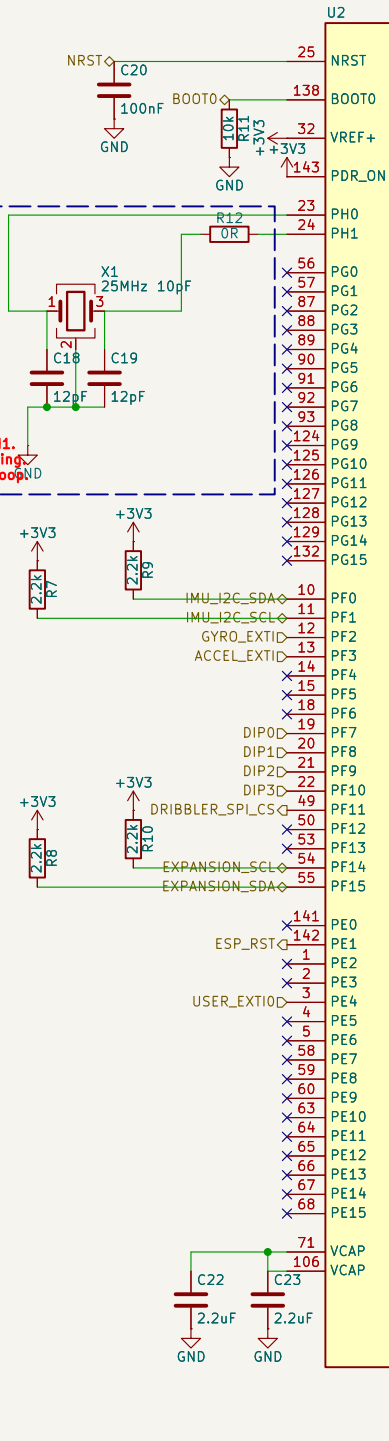
VDD DECOUPLING (EACH CAP AT ITS PIN)

ROUTING – DECOUPLING: each cap beside the STM32 pin named under it, via straight to GND.
C22/C23 (VCAP) tight to p71/p106. The netlist does not keep this pairing.



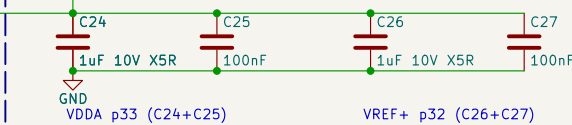
HSE 25MHZ CRYSTAL

ROUTING – HSE CRYSTAL
X1, C18, C19 within 5mm of PH0/PH1.
Short equal OSC traces, GND guard ring.
Nothing routed under or beside the loop.

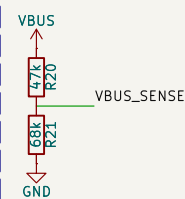


VDDA / VREF+ DECOUPLING

VDDA p33 and VREF+ p32 use +3V3.
Separate local bypass pairs; analog noise must be measured.
Source: DS13313 supply scheme / electrical characteristics.



VBUS SENSE



R15/R16 = 30R; provisional source damping, not certified matching.
No fixed driver resistance or SSR target follows from DS13313 Tables 52/54.
Validate SCK/MOSI/MISO/CS at every cable receiver, both directions.

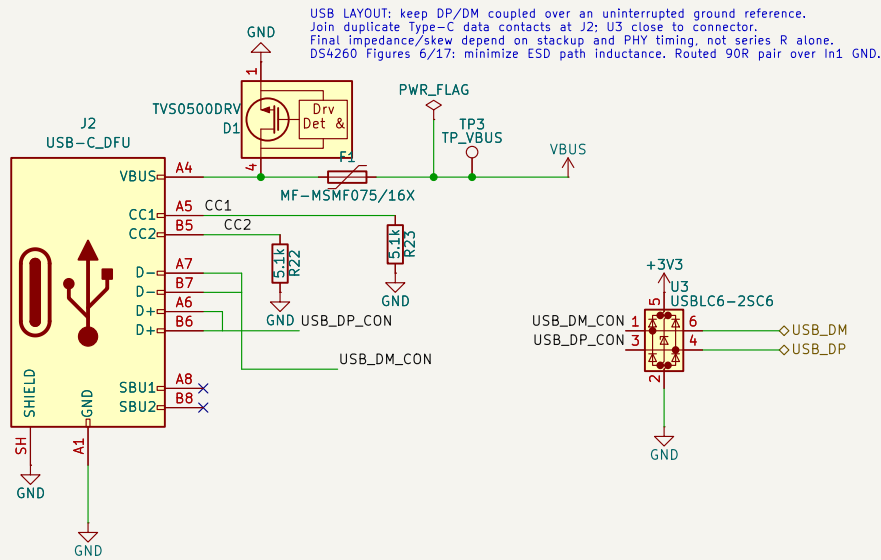
ROUTING – MOTOR SPI (6 Mbit/s, 5 drops)
R15/R16 within 5mm of PA5/PA7, not at the connectors.

I2C: R7/R9, R8/R10, R14/R17, R18/R19 are each 2.2k.
At 3.3 V, (3.3-0.4)/2.2k = 1.32 mA sink requirement.
For a 300 ns rise-time limit: C <= 300ns/(0.8475R) = 161 pF nominal.
At +5% R: 153 pF; include ESD pins, cables and parallel pull-ups.
LSM6DSGX20X / SSD1306 / DS13313 timing and I/O data; external loads not qualified.

I2C VALIDATION REQUIRED: measure SDA and SCL at worst load.
Do not change to 1.5k from rise time alone: recheck every device's VOL/current.
OLED module pull-ups and expansion-device specifications are not established.

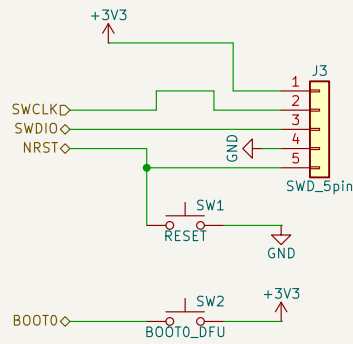
CLOCK: X1 = 25 Mhz; set firmware clock tree to the fitted part.
R12 = 0R is an oscillator drive-tuning link, not line termination.
C18/C19 need measured stray capacitance to establish the 10 pF load.
Startup margin and crystal drive remain unverified (ABM8, DS13313 HSE data).
No blanket claim that 25 Mhz always starts or HSI48 is unusable.
SUPPLIES: PDR_ON and VBAT are on +3V3; no backup battery.
VCAP 71/106 share their bypass node; retained pending ERC symbol review.
Do not separate a required supply node merely to silence ERC.
Source: DS13313 power-supply scheme / VCAP data.

USB-C J2: TVS, FUSE, CC RESISTORS, ESD



USB LOAD: MF-MSMF075/16X Ihold = 0.75 A at 23C, 0.55 A at 50C.
Actual concurrent worst-case board/external current is not established.
Restricted bench use until host current, startup and thermal budget are checked.
F1 does not negotiate USB current; replacing it alone cannot approve more load.

SWD DEBUG J3, RESET AND BOOT BUTTONS



SWD is direct; damping depends on probe/cable edges and layout.
DS13313 does not prescribe a universal 22R or 55R series resistor.
NRST has an internal 30-50k pull-up; C20 = 100nF (DS13313 NRST data).
U3 clamp rail is +3V3; validate rail injection and power-off behavior.
USBLC6-2 DS4260 pinout/topology: short return path required.

Sheet: /USB and Debug/
File: 03_USB_Debug.kicad_sch

Title:

Size: A3

Date:

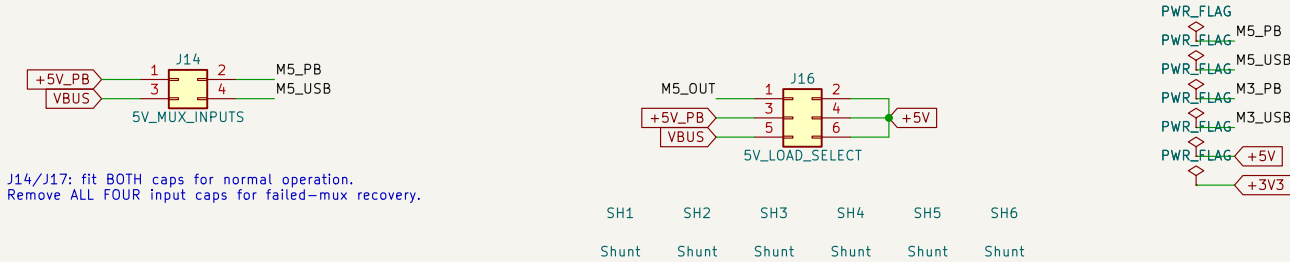
Rev:

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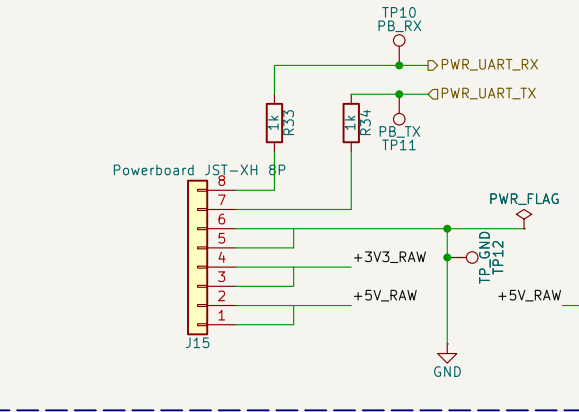
Id: 4/10

5V POWER PATHS: J14 INPUT LINKS / J16 LOAD SELECT

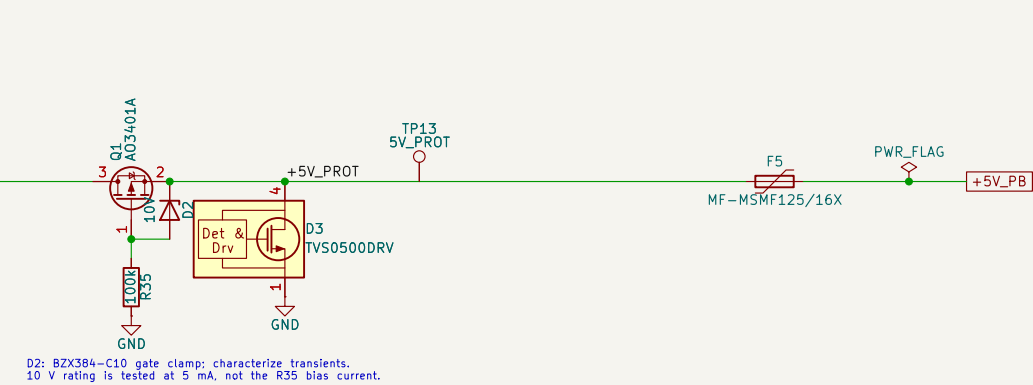
3V3 POWER PATHS: J17 INPUT LINKS / J18 LOAD SELECT



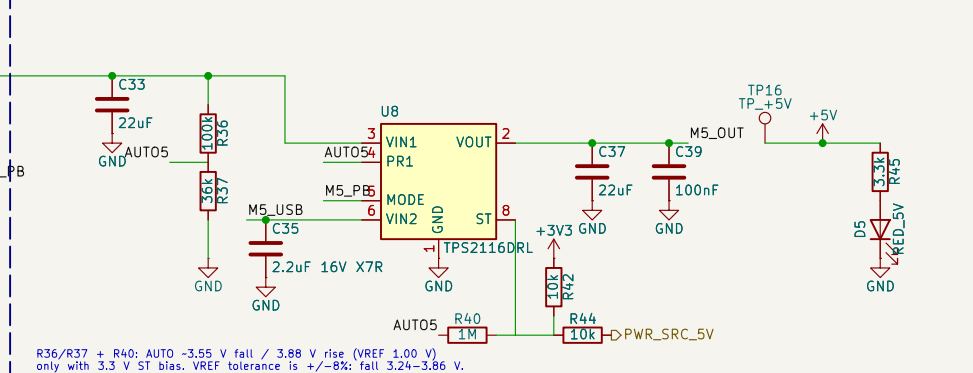
POWERBOARD CONNECTOR J15



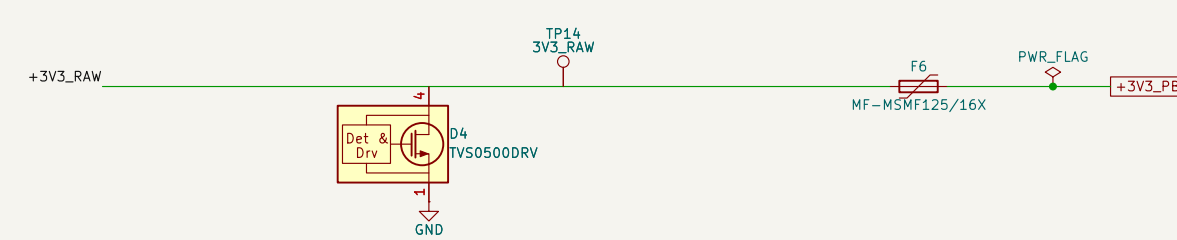
5V INPUT: Q1 REVERSE POLARITY, D3 TVS, F5 FUSE



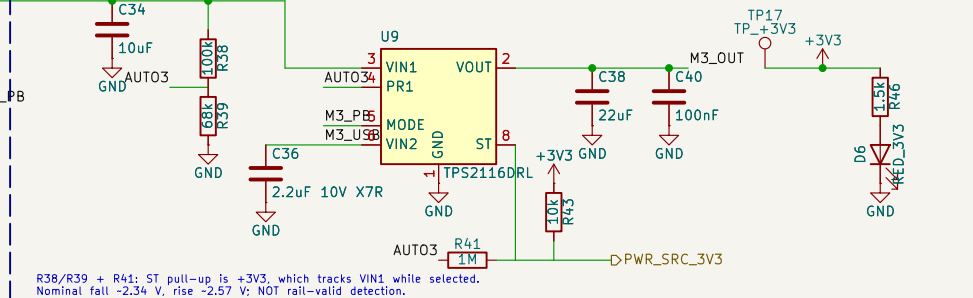
5V MUX U8: AUTO; BYPASS AT J16



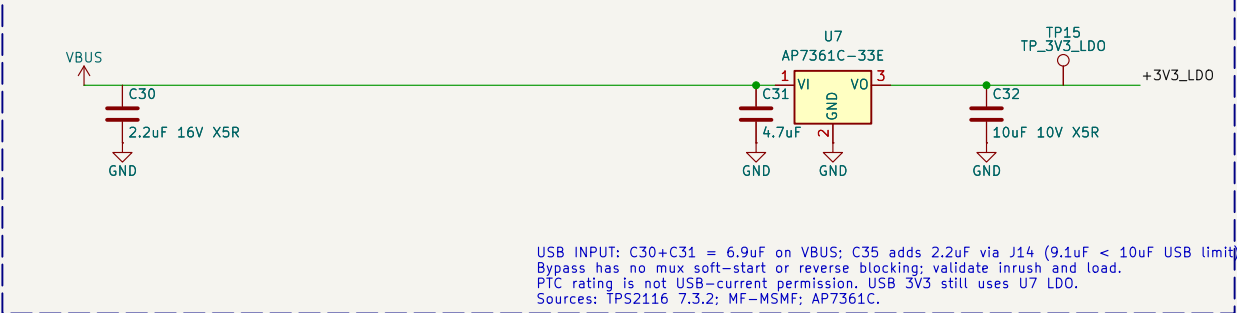
3V3 INPUT: D4 TVS, F6 FUSE



3V3 MUX U9: AUTO; BYPASS AT J18



USB VBUS AND 3V3 LDO U7



PHYSICAL BYPASS: J16/J18 carry load current, not PR1 control.
For recovery: power OFF, remove all J14/J17 caps, select PB or USB on J16/J18.
Bypass omits mux reverse blocking / soft-start. Source fuses remain.
ST reports MUX state only; cannot identify bypass source or USB current budget.
Control/status resistors remain; NOT galvanic or arbitrary-fault isolation.
Sources: TPS2116 SLVSFG1A 7.3 / 7.6.1; Samtec TSW and SNT datasheets.

J15: 1-2 +5V; 3-4 +3V3; 5-6 GND; 7 TX; 8 RX.
Mate/cable pinout and load remain external-interface requirements.
Q1 polarity agrees with AO3401A pinout. No 3V3 reverse-polarity FET.
R33/R34 limit current; they do not guarantee powered-off isolation.
PTC faults and mux source change can still interrupt other loads.

PROTECTION LIMITS: TVS0500 clamps up to 9.2 V at 43 A.
This is NOT proof of protection for 6 V-absolute-max TPS2116 or 3V3 loads.
PTCs are thermal devices, not fast current limits; verify fault coordination.
Sources: TVS0500, TPS2116 absolute ratings, Bourns MF-MSMF.

Sheet: /Power/
File: 01_Power.kicad_sch

Title:

Size: A3

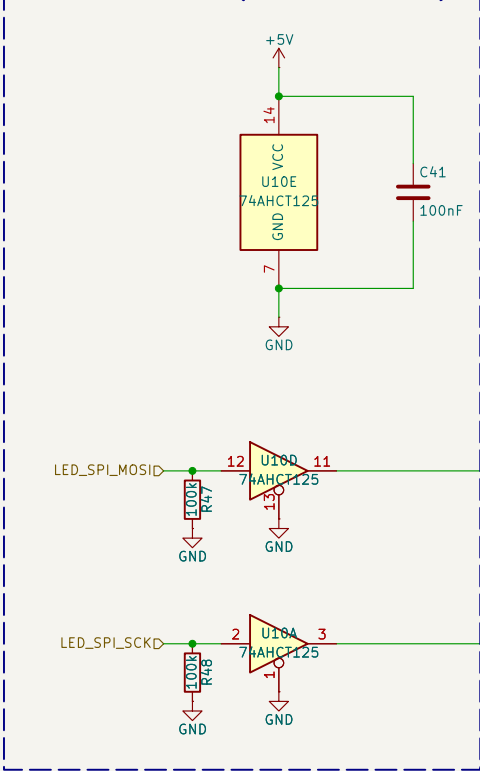
Date:

Rev:

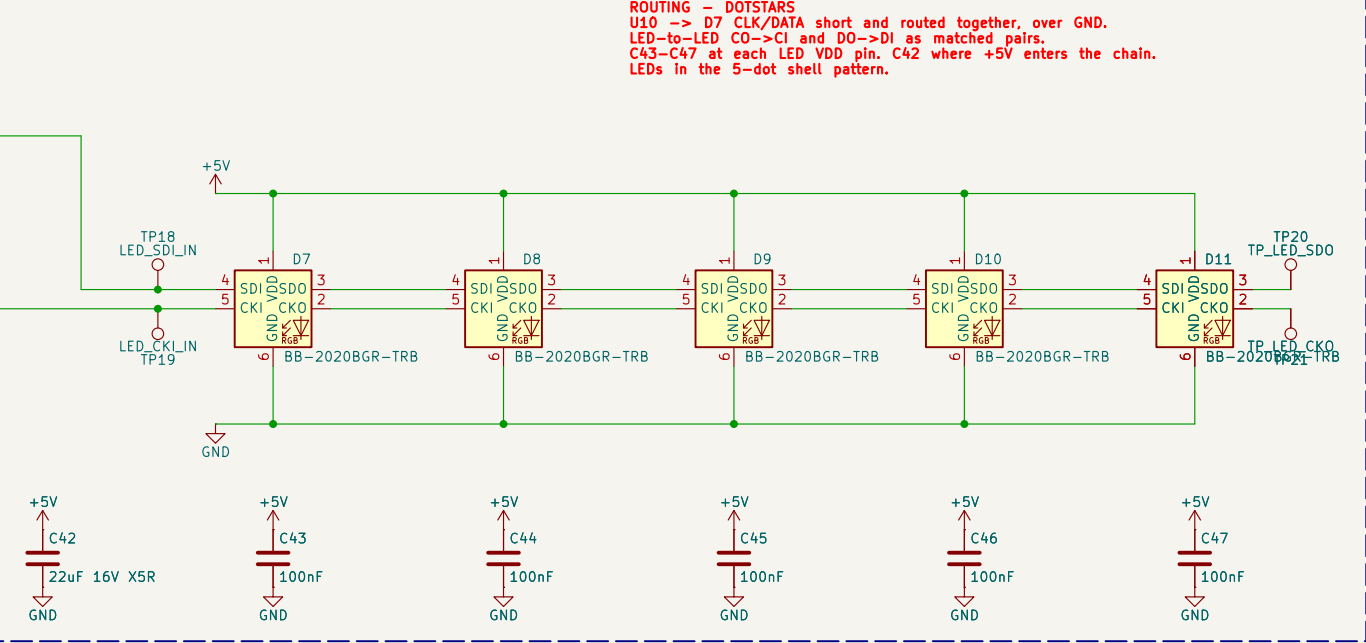
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Id: 6/10

LEVEL SHIFT U10 (3.3V SPI TO 5V)

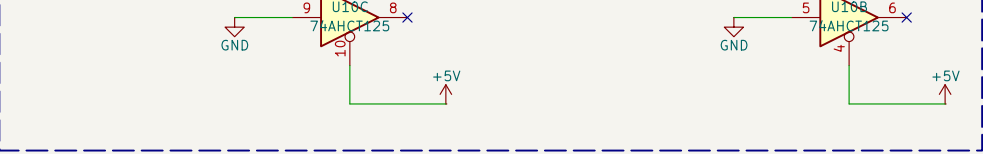


DOTSTAR CHAIN D7-D11 AND DECOUPLING



ROUTING - DOTSTARS
U10 -> D7 CLK/DATA short and routed together, over GND.
LED-to-LED CO->CI and DO->DI as matched pairs.
C43-C47 at each LED VDD pin. C42 where +5V enters the chain.
LEDs in the 5-dot shell pattern.

U10 SPARE GATES



LED LEVEL SHIFT: U10 = SN74AHCT125, specified for VCC 4.5-5.5 V.
3.3 V logic satisfies TTL input levels at a valid U10 supply.
R47/R48 = 100k bias inputs low; unused gates disabled.
D7-D11 = American Bright BB-2020BGR-TRB; use its data sheet/pad drawing.
A muxed USB source does NOT guarantee >=4.5 V after all cable/fuse drops.
Firmware current caps are requirements, not evidence of implemented limits.
Verify LED startup/current, power budget and custom footprints.
TP18/19 check chain input; TP20/21 check chain output.

Sheet: /DotStar LEDs/
File: 06_LEDs.kicad_sch

Title:

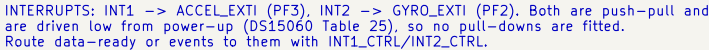
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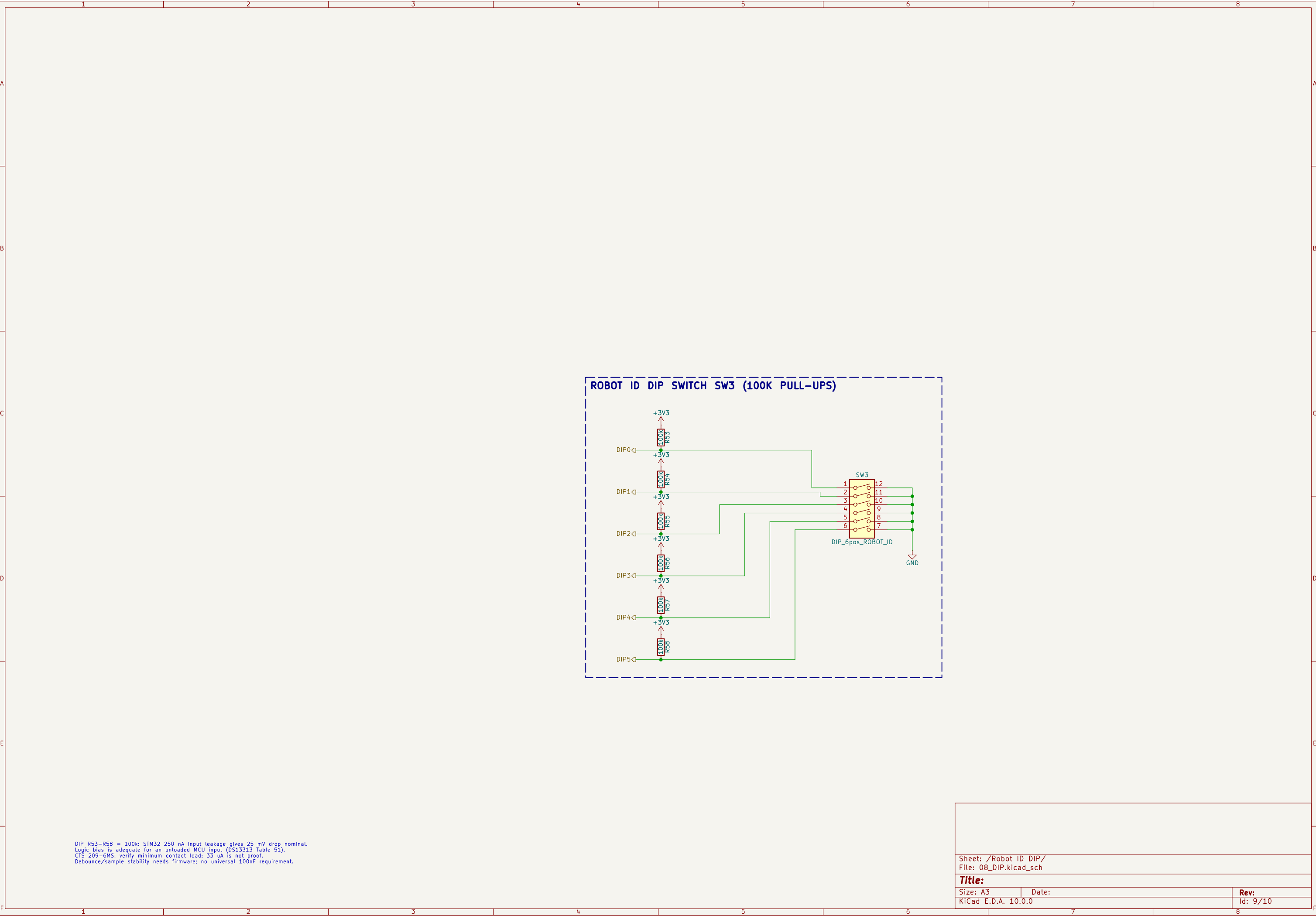
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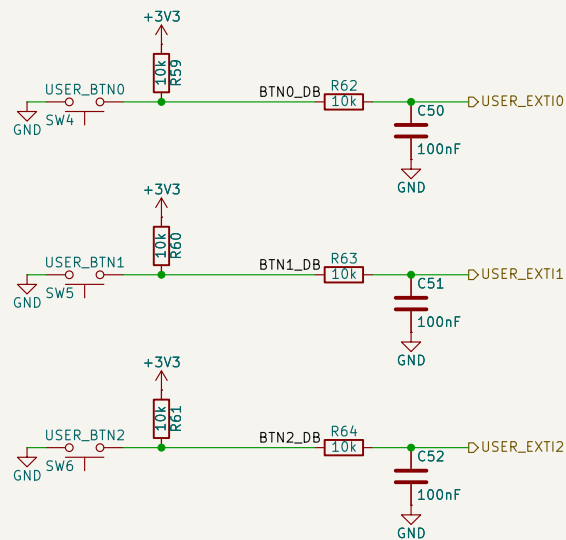
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Id: 7/10





USER BUTTONS SW4-SW6 WITH RC DEBOUNCE



BUTTONS: R59-R61 pull up with 10k; R62-R64 series 10k; C50-C52 = 100nF.
Calculated nominal time constants: press 1 ms, release 2 ms.
At +/-1% R and +/-10% C: 0.891-1.111 ms and 1.782-2.222 ms.
BOM: R59-R64 are 1%; C50-C52 need a +/-10% X7R part.
RC filtering does not guarantee removal of all bounce; quality switch and firmware.
Series R is between switch node and MCU/cap node, not between cap and MCU.
Input leakage check uses STM32H723 DS13313 Table 51.